

CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2016

BIOLOGY
Higher 2

Paper 3

19th SEP 2016/ Monday
2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

At the end of examination,

1. fasten all your work securely together

FOR EXAMINER'S USE	
1	/14
2	/14
3	/12
	/40
4	/12
5	/20
TOTAL	/72

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 14 printed pages
Answer **all** questions

Question 1

The first ever gene therapy trial was initiated in 1990 by Dr William French Anderson to treat a four year old girl named Ashanthi. Ashanthi was suffering from severe combined immunodeficiency (SCID), a genetic disorder characterised by the absence of functional T-lymphocytes.

In Ashanthi's case, the disease was caused by the absence of the enzyme adenosine deaminase (ADA-SCID). An alternative form of SCID is known as X-linked SCID.

- (a)** From your knowledge, contrast between the two forms of SCID mentioned in the paragraph above. [3]

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There were several *ex vivo* methods of gene therapy that were considered in Ashanthi's treatment, detailed in table 1.1. The virus vector used was a modified retrovirus.

Method	Description
A	• Normal ADA allele is introduced into viral vector • Recombinant virus is introduced into T-cells obtained from patient • Genetically modified cells are reintroduced into patient
B	• Normal ADA allele is introduced into viral vector • Recombinant virus is introduced into the cells derived from inner cell mass of blastocyst • Genetically modified cells are reintroduced into patient.
C	• Normal ADA allele is introduced into viral vector • Recombinant virus introduced into hematopoietic stem cells obtained from patient • Genetically modified cells are reintroduced into patient

Table 1.1

(b) With reference to the information presented in Table 1.1,

(i) Suggest one reason why *ex vivo* approach was utilised for gene therapy. [1]

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(ii) State the most preferred method of *ex vivo* gene therapy for ADA-SCID and justify the preference over the other 2 methods. [3]

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(c) Explain why the retrovirus was an efficient vector in the gene therapy of Ashanthi's condition. [2]

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Apart from diseases that plague human health, the effect of plant diseases on agriculture have also been in the spotlight. One such disease is the ringspot virus that plagues the papaya agricultural industry. Scientists have developed effective circumventive methods to tackle the problem of the ringspot virus through genetically modifying papaya. To do this, viral genes encoding capsid proteins were transferred to the papaya genome. These viral capsid proteins elicit something similar to an “immune response” from the papaya plant. Thus, the genetically modified papaya plants were resistant to infection by the papaya ringspot virus.

Figure 1.2 below depicts the comparative infection of transgenic and non-transgenic papaya in the 1995 field trail in Kapoho, Hawaii.

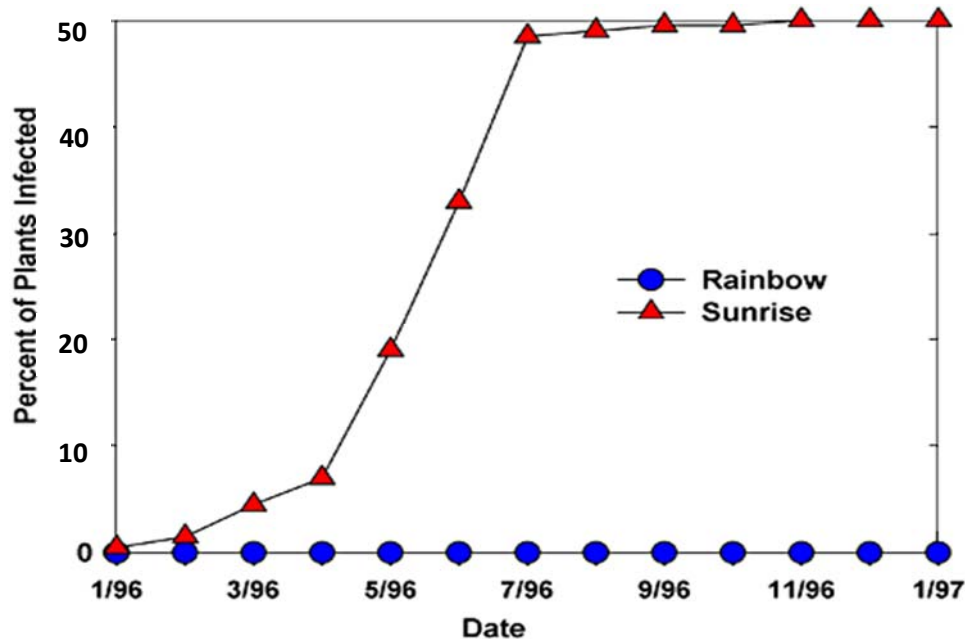


Figure 1.2

(d) With reference to Figure 1.2,

(i) Determine the identities of the transgenic and non-transgenic papaya species. [1]

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(ii) Evaluate and justify thoroughly the efficacy of the genetic intervention. [2]

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It was observed that three years later, the percentage of infected transgenic papaya species increased.

(e) Suggest and explain a possible reason for this phenomenon. [2]

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[Total: 14 marks]

Question 2

The location of the gene locus responsible for disease X was not discovered until 1985. Scientists used restriction fragment length polymorphism to discover the genetic markers associated with the disease. One such marker was a 950bp-long region known as XD15 that had been sequenced prior to 1985.

Samples of DNA were obtained from a family known to have the condition. The XD15 locus was amplified by polymerase chain reaction and mixed with *Pst*I and *Eco*RI in two separate restriction digests. The results of gel electrophoresis followed by southern blot of both restriction digests are shown in Figure. 2.1.

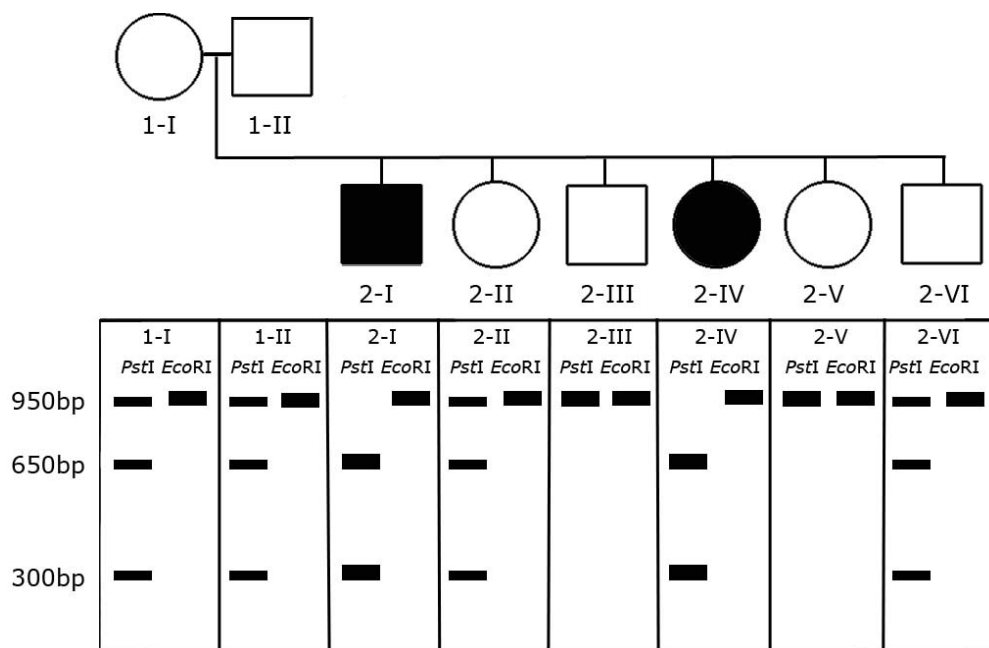


Figure. 2.1

(a) Using the information in Figure. 2.1,

- (i) state and explain which restriction enzyme digest should be used to detect the XD15 genetic marker associated with disease X. [3]

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(ii) Draw a restriction map of the XD15 genetic marker that is associated with disease X.

Indicate on the restriction map the position of the radioactively-tagged probe that would enable visualization of the RFLP fragments. [2]

(b) Explain why genetic markers like XD15 can be used to detect the presence of disease-causing alleles. [2]

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The XD15 locus was amplified by polymerase chain reaction prior to gel electrophoresis and southern blot. The DNA sequence of the XD15 locus is shown in Figure. 2.2.

5' - GGATCCATCCCGATCGAAAGCTAGCTAGGATCC - 3'
3' - CCTAGGTAGGGCTAGCTTTTCGATCGATCCTAGG - 5'

Figure. 2.2

(c) Design two 7-base long primers for the sequence to be amplified. [2]

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(d) Contrast between the process of PCR and DNA replication that occurs naturally in cells.
[3]

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(e) Besides disease detection, RFLP analysis may be used for DNA fingerprinting as well.
Explain one difference in the approach employed during RFLP analysis for both
processes (disease detection and DNA fingerprinting). [2]

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[Total: 14 marks]

Question 3

(a) Plant growth regulators (PGR) such as auxin, cytokinin and ethylene are naturally produced in plants and are important in determination of the developmental pathway of plant cells.

Synthetic analogues that can be mass produced are frequently used in plant tissue culture instead of the natural PGR.

Table 3.1 summarizes the general characteristics and roles of some natural and synthetic PGR.

Type of PGR	Examples	Role
Auxins	IAA(natural) - unstable to heat & light	Promote cell division and cell growth Root initiation (when auxin:cytokinin is high)
	2,4-D(synthetic) – stable to heat & light	
Cytokinins	2iP(natural) – unstable to heat & light	Promotes cell division Shoot formation (when auxin:cytokinin is low)
	Kinetin(synthetic) - stable to heat & light	
Abscisic acid (ABA)	-	Inhibits cell division Maturation of somatic embryos Abscission (i.e. shedding) of plant leaves Seed dormancy Induces stomatal closure to reduce water loss by transpiration
Ethylene	-	Abscission (i.e. shedding) of leaves and flowers Seed and bud dormancy by growth inhibition Fruit ripening

Table 3.1

Prunus lannesiana is an early-flowering cherry (only in spring) in Japan Izu peninsula. It is commonly known as Sakura or Japanese Cherry. Researchers are keen to propagate *P. lannesiana* from sterilized explants by micropropagation due to the advantages that the technique offers. Hence, a study was made to analyse the concentration of PGR present in the *P. lannesiana* in the four seasons and the results were summarized in Table 3.2.

Conc. of PGR / arbitrary unit \ Season	Spring (Mar-May)	Summer (June – Aug)	Autumn (Sep-Nov)	Winter (Dec – Feb)
IAA (auxin)	15	20	25	30
2iP (cytokinin)	10	15	10	5
ABA	5	5	12	15
Ethylene	12 – 20	18	16	14

Table 3.2

Table 3.3 illustrates the physiological development of *P. lannesiana* in the various seasons.

Season	Spring	Summer	Autumn	Winter
Shoot development	Very Active	Active	Minimal	Nil
Root development	Minimal	Moderate	Active	Very Active
Leaves development	Very Active	Active	Senescence (leaves turning autumn yellow) and abscission	Nil
Flowers development	Short full bloom in early spring, followed by senescence	Nil	Nil	Buds develop but dormant in late winter
Fruit and seed development	Fruit and seed development in late spring	Seed maturation	Seed dormant	Seed dormant

Table 3.3

(i) State one economical limitation of micropropagation. [1]

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- (ii)** Suggest why synthetic PGR (e.g. 2,4-D and Kinetin) are used instead of natural PGR (e.g. IAA and 2iP) in plant tissue culture. [1]

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- (iii)** Using the information provided in Table 3.2 and Table 3.3, explain the effect of the change in the auxin:cytokinin ratio from Spring to Winter and vice versa. [4]

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- (b)** The Salmon Genome Project (SGP) is developed to increase knowledge of the biology of Atlantic salmon and aid agricultural breeding of the fish.

- (i)** Describe how the SGP serves to increase knowledge of the biology of Atlantic salmon. [2]

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In order to produce transgenic salmon expressing salmon growth hormone, sGH, the coding sequence of sGH gene is isolated from a library and cloned, using Sac I, into the plasmid expression vector, pBluescript II SK. The plasmid map is shown in Figure. 3.1.

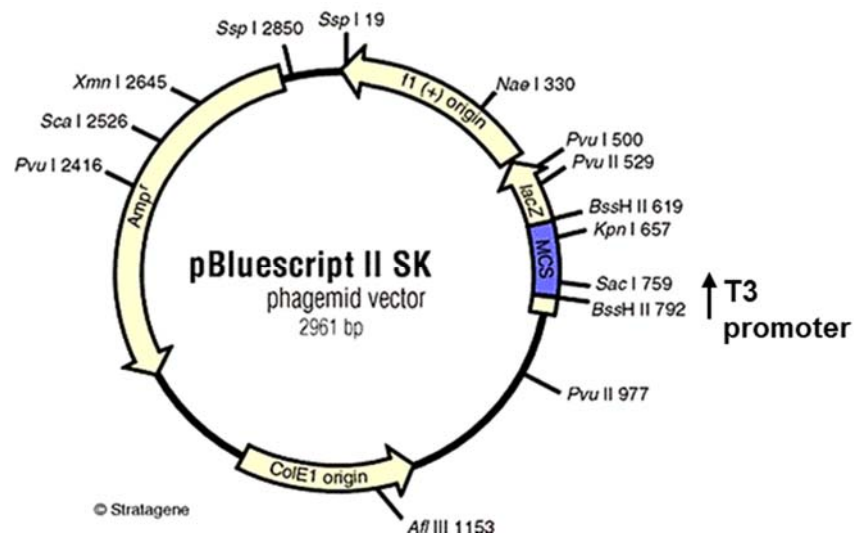


Figure 3.1

- (ii) With reference to figure 3.1, describe the features of the multiple cloning site (MCS). [2]

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Subsequent to insertion of sGH gene into pBlueScript II SK, transformation into *E. coli* cells was carried out and some colonies were obtained. The plasmid DNA was extracted, digested with Sac I and the restriction fragments were separated in gel electrophoresis.

- (iii) It was found that the recombinant plasmid with sGH gene inserted yielded no polypeptide. With reference to Fig. 3.1, state and explain a reason for this. [2]

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[Total: 12 marks]

Planning Question - Write your answers on the separate answer paper provided.

Question 4

Cefazolin is an antibiotic that disrupts the synthesis of bacterial peptidoglycan cell by preventing the formation of peptide bonds. It is bactericidal (kills bacteria) and is effective against gram-positive bacteria.

Streptococcus pneumoniae is a bacterium responsible for conditions like pneumonia and bacterial meningitis. It is a gram-positive bacterium which establishes itself as small white colonies. Discs containing cefazolin can be placed on an agar plate containing *Streptococcus pneumoniae*. If the cefazolin has been effective against *Streptococcus pneumoniae*, a clear zone will be seen around the disc as shown in Figure 4.1.

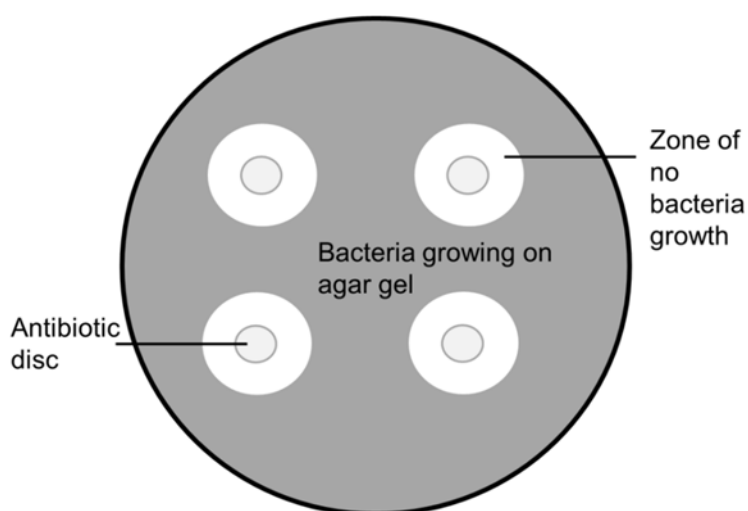


Figure 4.1

You are to plan but not carry out an experiment to investigate the effectiveness of different concentrations of cefazolin on the growth of *Streptococcus pneumoniae*.

Your plan must be based on the assumption that you have been provided with the following equipment and materials:

- Bunsen burner, to enable good aseptic conditions
- Bacterial culture in nutrient broth
- Molten nutrient agar
- Distilled water
- Sterile 90mm Petri dishes
- Sterile loops (to plate bacteria onto nutrient agar)
- 1cm³ pipette
- Filter paper discs
- Forceps
- Vernier calipers
- 1% cefazolin solution
- Bactericidal disinfectant for containment of used forceps and pipettes, also to clean work surfaces

Your plan should include:

- a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- an explanation of theory to support your practical procedure
- an explanation of the dependent and independent variables involved
- relevant, clearly labeled diagrams, if necessary
- proposed layout of results tables and graphs with clear headings and labels
- correct use of scientific and technical terms
- safety measures to minimise any risks associated with the proposed experiment

[Total: 12 marks]

Free-response question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 5

- (a) Discuss the advantages and evolutionary consequences of using plant tissue culture to propagate transgenic plants. [8]
- (b) Outline the ethical and social implications of genetically modified organisms. [6]
- (c) Explain why gene therapy is opined to have limited success in the treatment of genetic diseases. [6]